CSE 320-04 Quiz 5 (Set A)

Marks - 15 Time - 25 min

Name	
ID	

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1. Consider the following set of codewords: **[1.5 + 1.5 = 3]**
   {10110, 00011, 11111, 01101}
   a. How many bits of **error detection** can we get if we use these codewords?
   b. Find how many bits of **error correction** we can get.

2. Consider the cyclic redundancy check for the following case: **[4 + 3 = 7]**
   Dataword: **11011**
   Divisor: **1011**
   a. Find the codeword calculated by the sender.
   b. Show the verification process by the receiver when the 0-th and 3-rd bits of the codeword are flipped when receiving.

3. You are given a set of data: {0111, 1010, 1100, 0101} **[3 + 2 = 5]**
   a. Find the checksum calculated by the sender.
   b. Find the verification process of the receiver when the 0-th bit of the checksum is flipped.